

**23rd International Conference on Manufacturing Research
2–4 September 2026
ICMR 2026 Program**

Day 1: Wednesday, 2 September 2026

08:30 – 09:15	Registration (C3 - Red Hall)
09:15 – 9:45	Welcome and Formal Opening of the Conference: Professor Waqar Ahmad, President Professor Rehan Sadiq, Provost Professor Bjorn Birgisson, Vice-Provost for Research and Innovation Professor Essam Shehab, Acting Dean and General Chair of ICMR 2026 Conference Photograph – All delegates (C3 – Red Hall)
9:45- 10:45	Plenary Session 1: (C3 – Red Hall) Professor Ronan Daly, Institute for Manufacturing, Cambridge University, UK “Exploring the Role of Academic Manufacturing Research in Delivering New Medical Technologies” Chair: Prof. Huck Ng
10:45–11:15	Coffee Break (C3 – Ground Floor)

11:15-12:30	Industry-Academia Panel Discussion (C3 – Red Hall) “Made in Kazakhstan: Can We Build a Manufacturing Nation?” Panelists: Mr Pierre Lutz (Alstom), Mr Aksel Schmidt (Air Liquide), Mr Shynggys Mukhamedkali (Wabtec), Prof. Bjorn Birgisson (Nazarbayev University), Prof. Yusuf Altintas (University of British Columbia, Canada) Moderator: Prof. Essam Shehab (Nazarbayev University)
12:30-13:30	Lunch (C2 - Ballroom Left)

13:30-14.30	Plenary Session 2: (C3 – Red Hall) Workshop: Smart Manufacturing Prof. Yusuf Altintas (University of British Columbia, Canada) Chair: Prof. Md. Hazrat Ali
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Parallel Session 1 (C2 - Ballroom Right)				
Time	Session 1A: Sustainable Systems in Manufacturing - 1 (Venue: C2 - Ballroom Right - A) Chair: Prof Kostas Salonitis	Session 1B: Manufacturing Ergonomics and Education (Venue: C2-Ballroom Right - B) Chair: Prof. Gulnur Kalimuldina	Session 1C: Asset Monitoring and Supply Chain (Venue: C2-Ballroom Right - C) Chair: Prof. Ahmed Maged	Session 1D: Additive and Hybrid Manufacturing - 1 (Venue: C2-Ballroom Right - D) Chair: Prof. Asma Perveen
14:30-14:50	Paper 8: Visual Transparency Signals in Sustainable Manufacturing: A Qualitative Analysis of Sustainability Imagery [Online] <i>Authors: Balaussa Shaimakhanova Azubayeva and Andrew Thomas</i>	Paper 20: Self-Powered Soft Robotic Exo-Glove with Integrated Triboelectric Tactile Sensors for Assistive Hand Function in Impaired Persons <i>Authors: Bekzat Azamatov, Ali Oryn, Shabnam Yavary and Gulnur Kalimuldina</i>	Paper 10: Design and Impact Mitigation Performance of Foam-Reinforced Multiwall Auxetic Tubular Structures <i>Authors: Benrose Prasad, Fethi Abbassi and Sherif Araby</i>	Paper 9: Comparative Study of Powder Feeding Behavior for Different Materials in Powder-Based Directed Energy Deposition (DED) <i>Authors: Yeldar Otyunshiyev, Alikhan Kalmakhanbet, Yi Nie, Christos Spitas and Boris Golman</i>

14:50-15:10	Paper 70: Integrated Biotechnological Manufacturing of Steviol Glycosides <i>Authors: Nurgul Nugumanova</i>	Paper 63: Safe and Sustainable by Design for Operator 5.0 Decision-Support: A Sociotechnical System Perspective Translated from Safety-Critical Healthcare <i>Authors: Firda Rahmadani and Mecit Can Emre Simsekler</i>	Paper 81: Risk-Driven Dynamic Supply Chain Reconfiguration in Industry 4.0: An Empirical SEM Analysis [Online] <i>Authors: Dasunika Ubeywana and Sameh Saad.</i>	Paper 39: Influence of vacuum on the mechanical properties and surface roughness of 3d printing assisted investment casting <i>Authors: Wajid Hussain, Muslim Mukhtarkhanov, Md. Hazrat Ali and Essam Shehab</i>
15:10-15:30	Paper 79: Towards a Staged Assessment Framework for Smart Monitoring in Sustainable Manufacturing <i>Authors: Ziyad Sherif, David Butler and Shoaib Sarfraz</i>	Paper 66: A Patient-Centered Digital Service-Production Framework for Telemedicine in Mental Health Settings <i>Authors: Firda Rahmadani, Ahmed Al Blooshi, Maha Al Mufti, Alaa Alqaryuti, Nadeen Faraj, Mecit Can Emre Simsekler and Khalifa Al Meqbaali</i>	Paper 47: Towards an integrated generative AI knowledge management system for energy-sector maintenance operations <i>Authors: Ali Alsayegh and Tariq Masood</i>	Paper 45: Cybersecurity in Additive Manufacturing: A Survey of Attack Vectors, Defense Mechanisms, and Future Directions <i>Authors: Muhammad Abdullah Niazi, Essam Shehab and Md.Hazrat Ali</i>

Parallel Sessions 2 (Ballroom Right)				
Time	Session 2A: Advanced Manufacturing Technologies (Venue: C2-Ballroom Right - A) Chair: Prof. Mozafar Saadat	Session 2B: Applications of Industry 4.0 (Venue: C2-Ballroom Right - B) Chair: Prof. Yerkın Abdildin	Session 2C: Applications of Machine Vision (Venue: C2-Ballroom Right - C) Chair: Prof. David Butler	Session 2D: Modelling and Simulation (Venue: C2-Ballroom Right - D) Chair: Prof. Amgad Salama
15:30-15:50	Paper 59: Manufacturability-Driven Redesign of a Wire Flame Spray Nozzle: Integrating Design for Manufacture and Computational Fluid Dynamics <i>Authors: Muhammed Muneem, Mozafar Saadat and Majid Malboubi</i>	Paper 3: Development of an Integrated Industrial IoT System for Real-Time Data Monitoring and Control <i>Authors: Zhalgas Bolatbayev</i>	Paper 12: Transformer-based Detectors (DETRs) for the Industrial Personal Protective Equipment (PPE) Detection <i>Authors: Fadhlan Hafizhelmi Kamaru Zaman, Syahrul Afzal Che Abdullah, Kok Mun Ng and Mohd Farid Md Salleh</i>	Paper 2: Assessing Operational Impacts of Recycling Strategies Using Stochastic Simulation [Online] <i>Authors: Ahmed Basager and Abdullah Alrabghi</i>

15:50-16:10	Paper 67: Leveraging Design for Manufacturing and Assembly and 3D Printing in Product Redesign <i>Authors: Elia Abdulkarim, Mozafar Saadat and Majid Malboubi</i>	Paper 80: DFMA and 3D Printing in Consumer Product Redesign: A Hair Dryer Case Study <i>Authors: Elia Abdulkarim, Mozafar Saadat and Majid Malboubi</i>	Paper 24: Automated inspection of returned MedTech devices using deep learning for computer vision [Online] <i>Authors: Divya Tiwari, Ioan Alexandru Herdea, Matthew Mannix, Yuri Mejia, Fiona Charnley and Ashutosh Tiwari</i>	Paper 53: Study of the influence of operating factors on the thermal operating mode of internal combustion engines <i>Authors: Aidarali Tulenov, Yermek Baubekov, Baurzhan Shoibekov, Gabit Bakyt and Gabit Bekbolatov</i>
16:10-16:30	Paper 68: Integrating Design for Manufacturing and Assembly with 3D Printing for the Optimization of a Wireless Headphone Assembly [Online] <i>Authors: Rana Meligy, Mozafar Saadat and Majid Malboubi</i>	Paper 31: A Conceptual Simulation-based Digital Twin Framework for Predictive Analysis and Optimisation of Operational Efficiency and Workforce Productivity [Online] <i>Authors: Ghayda Diabat, Ian Pye, Rasoul Khandan, James Gao, Mahrukh Muzaffar, Van Dang Le, Chi An Le, Nguyen Danh Nguyen, Jamaluddin Mahmud and Chi Hieu Le.</i>	Paper 78: Real-time dimensional measurement of hot forged components: a pixel-level analysis <i>Authors: Muthair Hafeez, Muftooh Siddiqi and David Butler</i>	Paper 50: Strategic Governance of Decentralised Autonomous Supply Chains: Macro-Environmental and Contextual Frictions [Online] <i>Authors: Funlade Sunmola, Patrick Burgess, Memunat Lami Salihu-Yusuf and Hakeem Omolade Sunmola.</i>
16:30-17:00	Coffee Break & Networking (Upper Atrium)			

Parallel Sessions 3 (Ballroom Right)				
Time	Session 3A: Remanufacturing - 1 (Venue: C2-Ballroom Right - A) Chair: Prof. Mozafar Saadat	Session 3B: Advanced Manufacturing Systems (Venue: C2-Ballroom Right - B) Chair: Prof Kostas Salonitis	Session 3C: Smart Manufacturing (Venue: C2-Ballroom Right - C) Chair: Prof. Didier Talamona	Session 3D: Remanufacturing - 2 (Venue: C2-Ballroom Right - D) Chair: Prof. Lee Than
17:00-17:20	Paper 25: Recycled Carbon Fiber in Circular Economy Applications: Recent Advances and Developments in Sustainable Composites [Online] <i>Authors: Faraj Alsubaie and Liu Yang</i>	Paper 21: Marjan-R1: Design, Fabrication, Thrust Allocation, and Initial Hydrodynamic Parameterization of a Custom Fully-Actuated ROV [Online] <i>Authors: Md Tarique Bin Hamid, Sarvat M. Ahmad, Masroor Khan, Syed Imtiaz Ali Shah and Ali Albeladi</i>	<u>Keynote Presentation 1</u> Title: “Including the Human Touch in Robotic Manufacturing and Industry 5 through Surrogate Process Digital Twinning”	Paper 55: Development of a hybrid oil heating system in conditions of a switched off diesel engine of a shunting diesel locomotive and analysis of its effectiveness <i>Authors: Gabit Bakyt, Malik Zhamankulov, Kuantkhan Sarsenov, Berikkhan Tleuzhanov, Irina Ashirbayeva and Aizhan Makhanova</i>

17:20-17:40	Paper 48: Agentic Generative AI for Aviation MRO Scheduling: A Capability-Gated Autonomy Framework [Online] <i>Authors: Maryam Busakhar and Ali Alsayegh</i>	Paper 43: A.C.T.I.V.E. — advanced cybernetic technology for intervention, versatility, and exploration: a rocker-bogie mecanum rover with Kalman-fused localisation, Bézier-curve navigation, and Limelight vision tracking <i>Authors: Targyn Faizulla and Rustam Askaruly</i>	<i>Speaker: Professor Ahmed Qureshi, University of Alberta, Canada.</i>	Paper 60: Manufacturing Capability Maturity Pathways: A Comparative Study of Emerging Industrial Clusters in Kyrgyzstan, Malaysia and Indonesia <i>Authors: Lee Lee Than</i>
17:40-18:00	Paper 75: Feasibility of flax/PLA thermoplastic composite for sustainable wind turbine blade repair <i>Authors: Edward Cant</i>	Paper 72: Design and Integration of a Crank-Angle-Synchronised In-Cylinder Gas Sampling System for Heavy-Duty Diesel Engine Combustion Diagnostics <i>Authors: Hameed Alshammari [Online]</i>	Paper 74: Development of a Low-Cost Intelligent Robotic Waste Sorting System Based on Computer Vision and Autonomous Manipulation <i>Authors: Dauren Baltabay</i>	Paper 4: Characterization and Modeling of Non-Spherical Granular Materials for High-Temperature Thermal Energy Storage <i>Authors: Aidana Boribayeva, Luis R. Rojas-Solórzano, Jennifer Curtis, Nicolín Govender and Boris Golman</i>

18:30–18:30	NU Laboratory Visit (C4)
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18:30–20:00	Conference Dinner (Ballroom Left)
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Day 2: Thursday, 3 September 2026	
09:00 – 09:30	Registration
09:30 – 10:30	Plenary Session 3: (C3 – Red Hall) Professor Yusuf Altintas, University of British Columbia, Canada “Mathematical Modelling of Machining Processes and CNC Machine Tools for Digital Twin Assisted Process Planning and Monitoring” Chair: Prof. Asma Parveen
10:30-11:30	Coffee Break (C3 – Ground Floor)

11.30-12:30	Plenary Session 4: (C3 – Red Hall) Professor Kostas Salonitis, University College Dublin, Ireland “Sustainability and Additive Manufacturing” Chair: Prof. Essam Shehab
12:30-14:00	Lunch Break (C2 - Ballroom Left)

Parallel Sessions 4 (C2 - Ballroom Right)				
Time	Session 4A: Digital Modelling and Monitoring (Venue: C2-Ballroom Right - A) Chair: Prof. David Butler	Session 4B: Supply Chain for Manufacturing (Venue: C2-Ballroom Right - B) Chair: Prof Yerkin Abdildin	Session 4C: Smart Systems (Venue: C2-Ballroom Right - C) Chair: Prof. Sherif Gouda	Session 4D: Digital Modelling and Monitoring (Venue: C2-Ballroom Right - D) Chair: Prof. Mozafar Saadat
14:00-14:20	Paper 32: Real-Time Localized Digital Twin for Quadrotor UAV Flight Monitoring <i>Authors: Azim Dukenderov, Meirzhan Mazhenov, Alikhan Duisenbay, Md. Hazrat Ali and Essam Shehab</i>	Paper 17: Effervescent Tablet-Based Nanofluid with Enhanced Thermal Characteristics and Neutron-Absorbing for Rapid Nuclear Reactors LOCA Response [Online] <i>Authors: Naser Ali and Fatima Safar</i>	<u>Keynote Presentation 2</u> Title: Intelligent Anomaly Detection in Manufacturing Systems Using Data-Driven Models	Paper 40: CommandNet for Target Pursuit: Learning-Based Control Evaluation [Online] <i>Authors: Almat Abdimalik, Nurtay Kaziyeu and Ilyas Muhammad</i>
14:20-14:40	Paper 38: Allowance-Constrained CAD-to-Point-Cloud Registration for Robotic Machining-Margin Analysis of Aircraft Panel Blanks <i>Authors: Yan Yang, Zhongwei Li, Kai Zhong and Shi Yusheng</i>	Paper 27: Methane Concentration Forecasting in Underground Mines Using Sensor Data: A Systematic Review of Methods, Datasets, and Open Challenges [Online] <i>Authors: Iskander Akhmetov and Bakyt Kurmanov</i>	<i>Speaker: Dr. Ahmed Maged American University Sharjah –UAE</i>	Paper 54: Urban Heat Island Dynamics Around Industrial Zones Using Landsat and MODIS Time Series <i>Authors: Shamma Albedwawi and Naeema Al Hosani</i>
14:40-15:00	Paper 57: Development of an Intelligent Control System for an Integrated Low-Carbon Energy System Based on FPGA, ESP32, MQTT Protocol, and a Digital Twin <i>Authors: Zhazira Shermantayeva</i>	Paper 1: Agile methodologies in procurement contracts for improving new product development in complex engineering systems. [Online] <i>Authors: Ayaulym Armanova</i>	Paper 23: Self-Powered Triboelectric Electronic Skin with Braille-Based Interface for Assistive Communication in Deaf-Blind Individuals <i>Authors: Raiymbek Nurtay, Olzhas Kanatbek, Shabnam Yavari and Gulnur Kalimuldina</i>	Paper 56: The Decoupled Information Boundary: Resolving the Capacity Truth Paradox in Distributed Autonomous Shared Manufacturing Organisations [Online] <i>Authors: Funlade Sunmola and George Baryannis</i>

Parallel Session 5 (C2 - Ballroom Right)				
Time	Session 5A: Artificial Intelligence and ML in Manufacturing (Venue: C2-Ballroom Right - A) Chair: Prof. Didier Talamona	Session 5B: Manufacturing Optimisation (Venue: C2-Ballroom Right - B) Chair: Prof. Altay Zhakatayev	Session 5C: Sustainable Systems in Manufacturing - 2 (Venue: C2-Ballroom Right - C) Chair: Prof. Kostas Salonitis	Session 5D: Artificial Intelligence and ML in Manufacturing (Venue: C2-Ballroom Right - D) Chair: Prof. Fadhlan Hafizhelmi Kamaru Zaman
15:00-15:20	Paper 30: Deep Learning-Based Segmentation and Morphological Characterization of Gamma-Prime Precipitates in Superalloy Microstructures [Online] <i>Authors: Isaac Iwediba and Vassili Vorontsov</i>	Paper 11: The Application of Process Mapping and Simulation for Optimising Productivity in Low-Volume Assembly Cells [Online] <i>Authors: Daniel Eaves, Andrew Rees, Andrew Thomas, Christian Griffiths and Balaussa Shaimakhan Azubayeva</i>	<u>Keynote Presentation 3</u> Title: Manufacturing of microfluidics-based lab-on-a-chip medical device for cell micromanipulation	Paper 19: From Prompts to Performance Rankings: Evaluating Large Language Models as Surrogates for Building Performance Simulation to Rank Residential Designs in Hot Climates <i>Authors: Abdalrahman Alajmi and Ali Alsayegh</i>
15:20-15:40	Paper 52: A Physics-Informed Machine Learning Framework for Defect Prediction in Directed Energy Deposition Using Simulation, Experimental Validation, and Generative AI [Online] <i>Authors: Yerlik Gabdulla</i>	Paper 62: Integrating digitalization into SMED for setup time reduction: A case study in corrugated board production <i>Authors: Tamer Abdulghani, Moritz Heyne and Herwig Winkler</i>	Speaker: Professor Mozafar Saadat, University of Birmingham, UK	Paper 37: Knowledge management in the age of generative AI: opportunities and challenges for Kuwait's energy sector <i>Authors: Abdalrahman Alajmi and Ali Alsayegh</i>
15:40-16:00	Paper 64: AI-driven voice biomarkers for early detection and continuous monitoring of schizophrenia <i>Authors: Raghad Al-Sakaji, Firda Rahmadani, Haya Al Jaghoub, Shayma Alshehhi, Sarah Alkindi, Faisal Nawaz and Mecit Can Emre Simsekler</i>	Paper 2: Assessing Operational Impacts of Recycling Strategies Using Stochastic Simulation [Online] <i>Authors: Ahmed Basager and Abdullah Alrabghi</i>	Paper 71: A geometric-kinematic feasibility framework for robotic cable clip installation [Online]	Paper 65: From Algorithmic Performance to Clinical Adoption: A Review-Based Taxonomy of Artificial Intelligence Clinical Decision Support Deployment Challenges <i>Authors: Ahmed Saad, Firda Rahmadani, Khaled Toffaha, Mecit Can Emre Simsekler and Khalid Alzarooni</i>
16:00-16:30	Coffee Break (Upper Atrium)			

Parallel Sessions 6 (C2 - Ballroom Right)				
Time	Session 6A: Machining Processes and Advanced Materials (Venue: C2-Ballroom Right - A) Chair: Prof Asma Perveen	Session 6B: Automation, Flexible Assembly, and Sensors (Venue: C2-Ballroom Right - B) Chair: Prof. Fadhlán Hafizhelmi Kamaru Zaman	Session 6C: Additive and Hybrid Manufacturing - 2 (Venue: C2-Ballroom Right - C) Chair: Prof. Yerbol Sarbassov	Session 6D: Machining Processes and Advanced Materials (Venue: C2-Ballroom Right - D) Chair: Prof. Sherif Gouda
16:30-16:50	Paper 13: A Finite Element Study of the Graphene Coated- Bamboo Fibres Reinforced Polyethylene Composites. [Online] <i>Authors: Faraj Alsubaie</i>	Paper 7: IoT-Enabled Predictive Maintenance for Industrial Automation Systems <i>Authors: Alisher Yelseitov, Zhalgas Bolatbayev and Tohid Alizadeh</i>	Paper 28: Rapid Liquid Printing in Clay Medium for Construction-scale Concrete 3D Printing: A Process Development <i>Authors: Mukhammad Safa Akimbay, Bakbergen Temirzakuly, Essam Shehab and Md. Hazrat Ali</i>	<u>Keynote Presentation 4</u> Title: Composite Materials: From Conventional to Advanced Manufacturing for Emerging Applications <i>Speaker: Professor Fethi Abbassi American University of the Middle East – Kuwait</i>
16:50-17:10	Paper 51: Integrated Thermodynamic Modeling and Process Optimization for Microstructure Control in LPBF-Processed CoCrNi–CuZr–Ta Medium-Entropy Alloys (MEAs) <i>Authors: Clifford Omonini</i>	Paper 42: Sensor-based physical safety system for human-robot collaboration <i>Authors: Bakbergen Yermekbayev, Kaisar Kozhay, Muhammad Abdullah Niazi, Md Hazrat Ali and Essam Shehab</i>	Paper 29: Geometry-aware inverse process planning for non-planar extrusion additive manufacturing <i>Authors: Kaicheng Ruan, Yiwei Weng and Yi Xiong</i>	
17:10-17:30	Paper 49: Development of Sustainable Aerated Geopolymer Concrete Using Steel Industry By-products <i>Authors: Kassym Talgatuly, Chang-Seon Shon, Nazerke Kabylbek, Aigerim Baizhigitova and Saken Sandybay</i>	Paper 14: Optimization of production rate of WEDM using Taguchi Methodology <i>Authors: Rashmi Arya and Hari Singh</i>	Paper 44: A comparative study of energy-efficient WEDM strategies for better surface integrity for machining of hybrid composite <i>Authors: Nilesh Kumar, Jatinder Kumar and Rajesh Khanna</i>	
18:00-21:00	Conference Gala Dinner (Ticket Holders Only) - (C2 - Ballroom Left) Award Ceremony and Closing Remarks			

Day 3: Friday, 4 September 2026 - City Visit	
08:30-09:00	Registration (Block 1 - Ground Floor)
09:00-12:30	City Visit